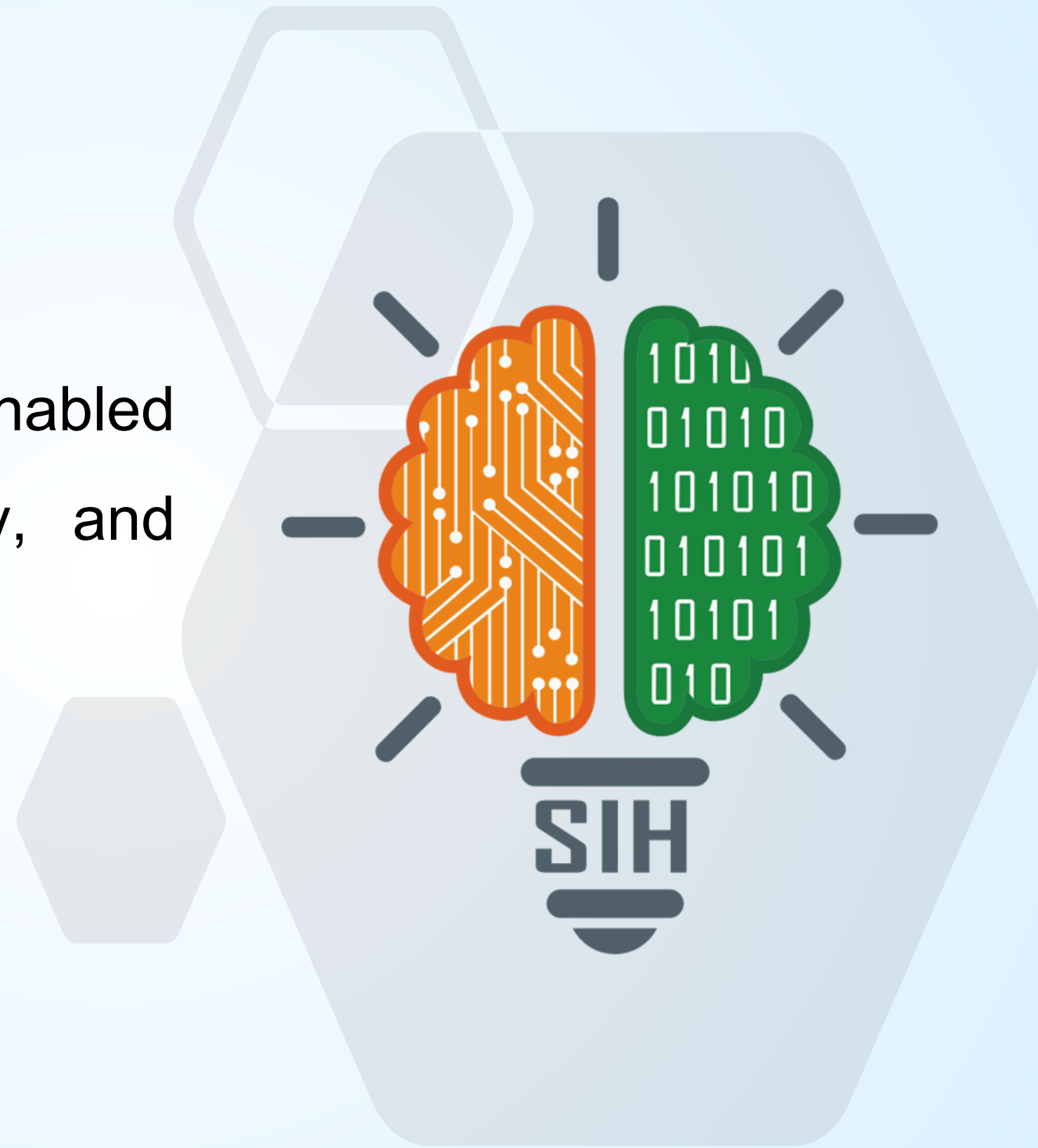


- **Problem Statement ID – 26059**
- **Problem Statement Title-** AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 93C4B2
- **Team Name-** Krisis

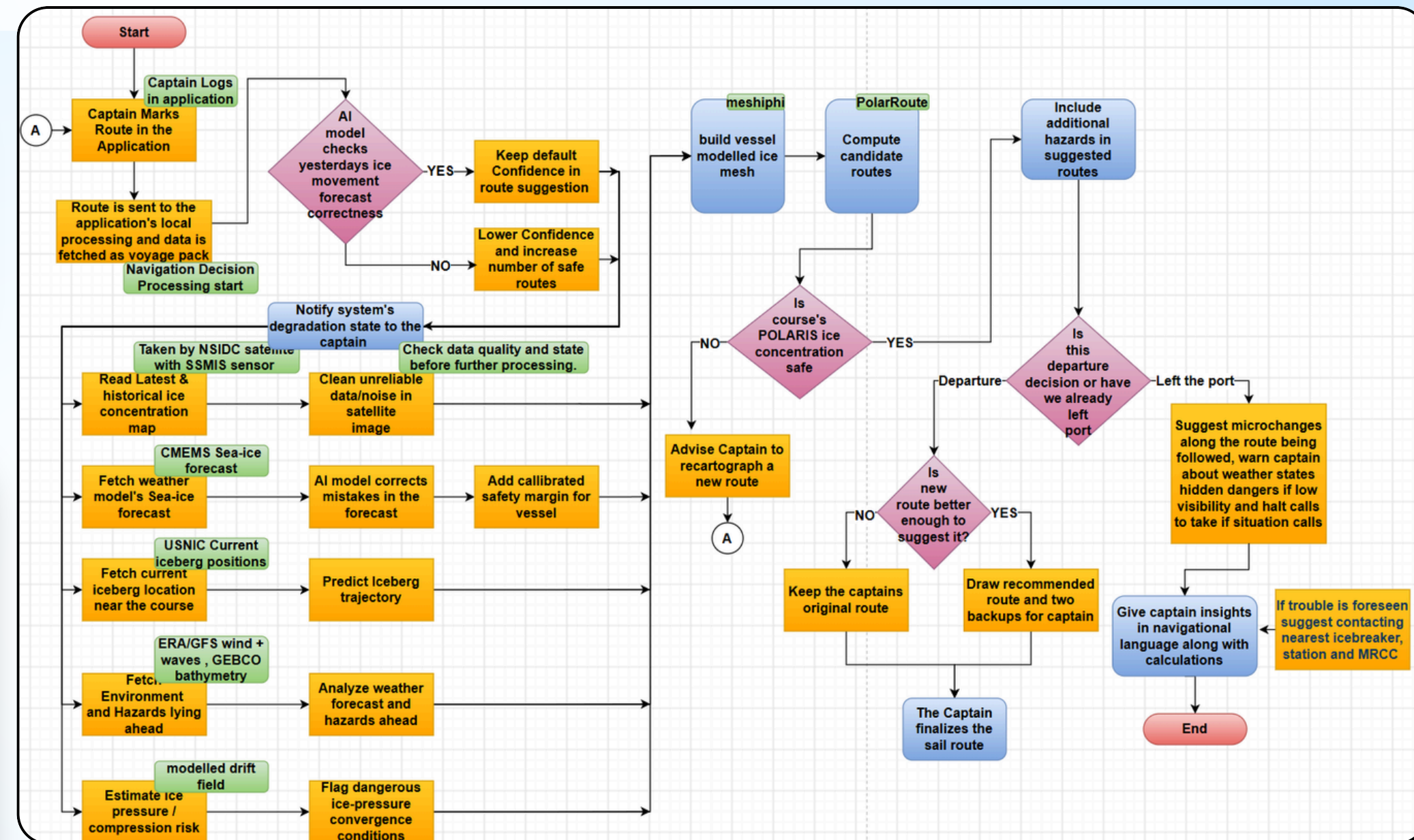


Solution Idea

Our solution aims to create an AI/ML-enabled decision-support platform that forecasts Antarctic sea ice, predicts iceberg trajectories, and identifies safe, fuel-efficient vessel routes using satellite, oceanographic, and meteorological data.

Proposed solution / Approach

- **Ice Forecast Correction:** AI corrects the official sea-ice forecast, improving accuracy by **18.5% at 1 day** and **30.6% at 7 days** over the baseline.
- **Iceberg Drift:** A proven physics model predicts the drift of **live icebergs**, including **within our shipping corridor**.
- **Safe Routing:** The router finds the safest path using real satellite ice data, **avoids protected areas**, and **re-plans automatically** as new data arrives.
- **Ashore and Aboard:** Heavy **processing** runs **ashore**; the ship receives one small daily update via **Iridium** and runs the **full router offline**.
- **Captain Decides:** The system **advises**, explains **what changed and why**, and shows the **fallback level** being used. The **captain** always makes the **final decision**.



INNOVATION & UNIQUENESS

ERROR CORRECTION
Finds and fixes errors in ice movement forecasts.

COASTAL BLIND SPOT
Detects satellite gaps that falsely mark open water.

BUILT FOR IRIDIUM
76 KB data pack • 796-byte route.

4 SAFETY NETS
4 fallback levels • active level shown.

DAILY SELF-CHECK
Compares yesterday's forecast with today's satellite data.

CORE PRINCIPLE (THE RULE)



Ashore handles heavy computation (GPU/Bandwidth), Aboard handles real-time questions, and Sync is a compressed environmental state, not a finished answer.



1. SHORE TIER (OPS CORE)

GPU & BANDWIDTH

- 1 Data Ingestion
- 2 QC Pass & Regridding
- 3 Feature Assembly: 14 Channels
- 4 Model Training (GPU/Kaggle)
- 5 Output: SIC Mean (μ) & Uncertainty (σ)
- 6 Fusion Layer: Disagreement Masking
- 7 Iceberg Drift: Wagner + GBM Residual
- 8 Mesh Generation: meshphi
- 9 Routing: PolarRoute Sweep
- 10 Package Build: Voyage Pack (Signed/Versioned)



Workstation (GPU)



SYNC

Voyage Pack
(Compressed
Environmental
State)

- ✓ Signed
- ✓ Versioned
- ✓ Optimized for Low Bandwidth



Iridium
(Low BW)



USB Sneakernet
(High BW)

2. VESSEL TIER (BRIDGE)

LAPTOP / BRIDGE

- 1 Receive Voyage Pack (USB / Iridium)
- 2 Load: Cached Data & Model Weights
- 3 Local Compute: Real-Time Inference
- 4 Update: Fresh Obs → Re-correct Forecast
- 5 Fusion: Disagreement Layer Update
- 6 Hazard Layer Update: Berg Exclusions
- 7 Route Re-plan: PolarRoute on Dynamic Mesh
- 8 UI Render: Route & Risk Overlay



TECH STACK

Data & Ingest

- Python 3.11
- copernicusmarine
- xarray
- netCDF4
- GitHub Actions
- Zarr

Bridge Console

- React
- Leaflet
- TypeScript
- Vite

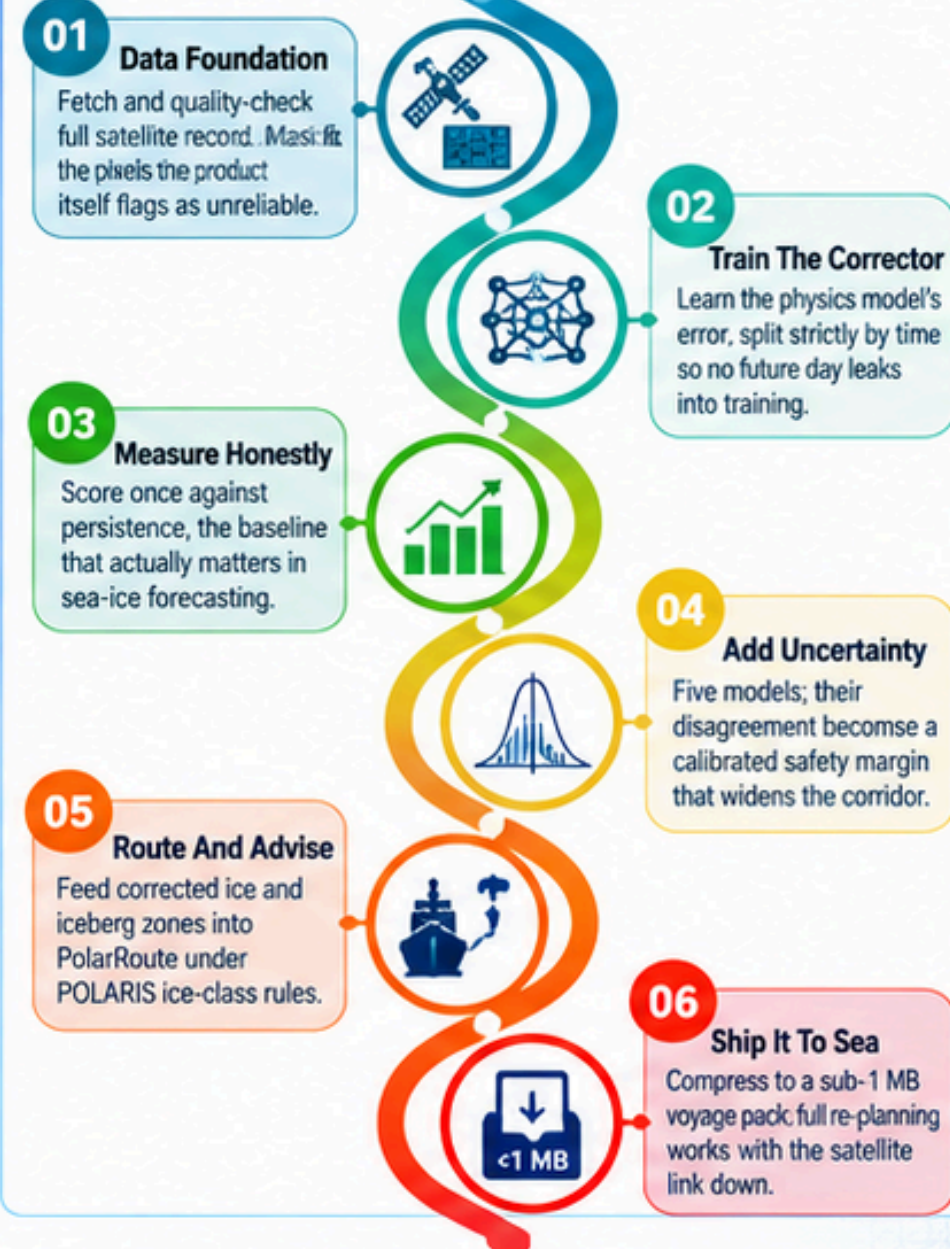
Backend

- FastAPI
- Uvicorn
- Pydantic
- Zstandard

AI & Routing

- PyTorch
- NumPy
- SciPy
- pandas
- pyproj
- PolarRoute
- meshphi
- pytest
- scikit-learn

Implementation Process



DATA SOURCES



NSIDC
Sea Ice Index



OSI SAF
Sea Ice (alt)



ESA
CCI



Sentinel-1
SAR Imagery



BYU
Iceberg Tracking



USNIC
Iceberg Tracking



NOAA
CMEMS / OISST



GFS
Weather & Waves



ECMWF
ERA5



GEBCO
Bathymetry



NCPOR Portal
(NPDC)

Feasibility:

- **Proven Model:** Beats the standard baseline by 18.5% at 1 day and 30.6% at 7 days on real held-out satellite data.
- **Real Pipeline:** 1,096 daily satellite files processed with zero failures, harvested automatically every day.
- **Reused Engines:** PolarRoute and meshphi from the British Antarctic Survey already compute our 8.6-day route.

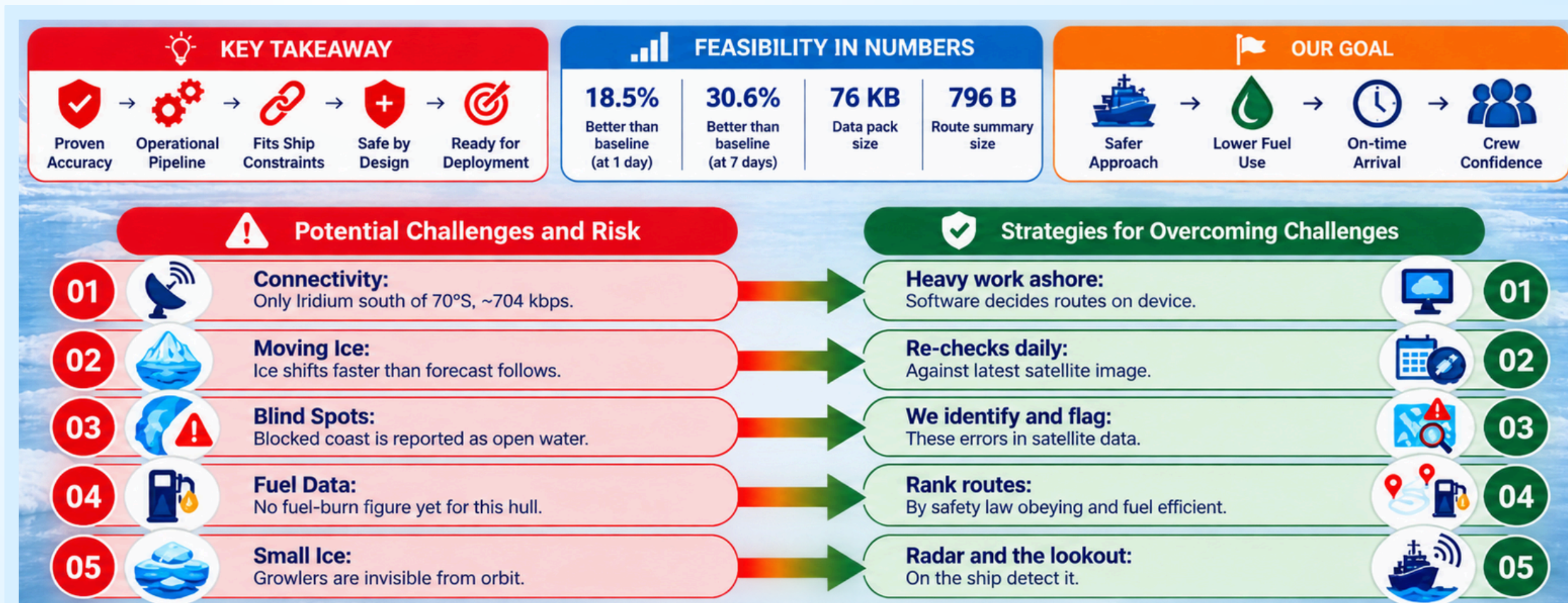
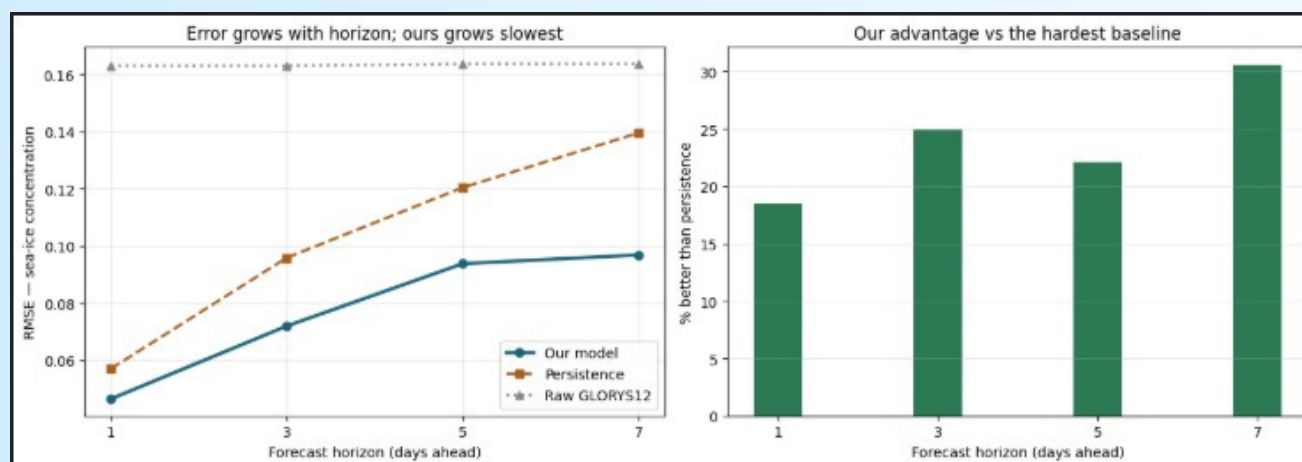
Viability:

- **Fits the Link:** A full data pack is 76 KB and a route summary is 796 bytes, well inside Iridium's 704 kbps.
- **Small Archive:** Compression cuts each daily file from 42.6 MB to 5.0 MB, 8.4x smaller.
- **Open Source:** Every tool in the stack is open source and the satellite data is public, with nothing to license.

Practical implementation:

- **Runs Offline:** The ship runs the full router locally with no internet; only a small daily update comes over Iridium.
- **Fails Safe:** The model outputs a correction to the official forecast, so if it fails the crew still has the official forecast, not nothing.
- **Already Running:** The daily data harvest started early and runs automatically, because missed days can never be back-filled.

Our Results:



Impact of the Solution



1. Ship's Crew: Sees where the ice will be, which icebergs are near, and why the route changed, with the captain always in command.



2. NCPOR: Plans the Cape Town to Bharati resupply on real ice forecasts and knows in advance when the Bharati approach is closed.



3. Scientists: Supplies and teams reach Bharati and Maitri on time, and our coastal data audit exposes a flaw in the satellite record they rely on.



4. The Nation: India gets a home-built decision tool for its Antarctic programme, built on open data and open code.



5. The Environment: The router stays inside the protected-area boundaries that India co-proposed, and safer routes cut the risk of grounding or spills.

Benefits of the Solution



1. Ship Safety First: The system explains what changed and why, giving the crew clear reasons for the route and keeping the captain in control at every step.



2. Economic: Fewer stuck days and smarter routes save costly ship time on a chartered vessel.



3. Danger Aware: Cautious routing through heavy ice lowers the risk of hull damage and spills in Antarctic waters.



4. Operational: Works with very little internet. Everything heavy runs ashore, the ship receives a small daily pack, and the bridge re-plans on a laptop in seconds.

01

**Ice Class Operating Limits (POLARIS):**

Replaced the initial 80% ice threshold with the POLARIS risk-index framework for ice-class operating limits [1].

02

**Iceberg Drift Model:**

Implemented the Analytical Model of Iceberg Drift, including correction of drag-coefficient inconsistencies [2].

03

**Sea-Ice Round Robin:**

Incorporated uncertainty reporting based on evidence that published sea-ice algorithms can differ significantly, especially near the ice edge [3].

04

**Sea-Ice Uncertainty:**

Calibrated uncertainty using documented producer exclusion criteria, including melt ponds, thin ice, and weather effects [4].

05

**Live Iceberg Positions:**

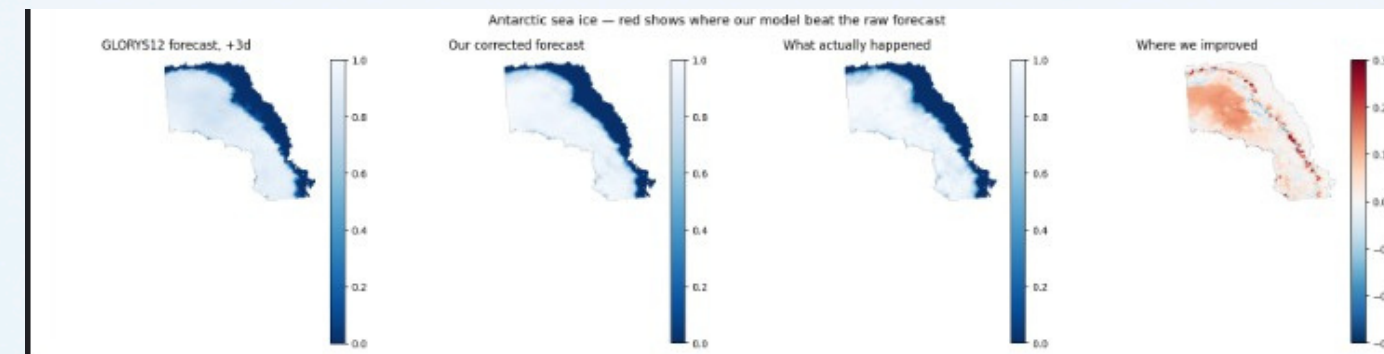
Used the U.S. National Ice Center (USNIC) Antarctic Iceberg Dataset for freely available, regularly updated iceberg positions [5].

06

**Additional Research:**

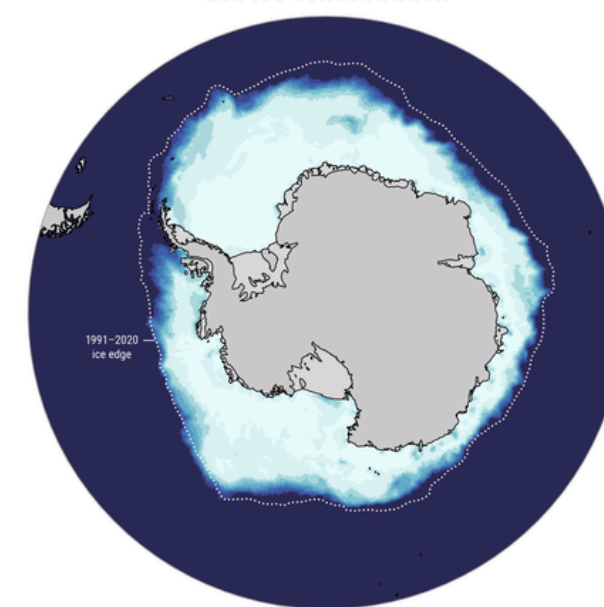
Reviewed Antarctic ship navigation, polar operations, navigation systems, and safety practices using multiple technical and operational sources.

Our Results:

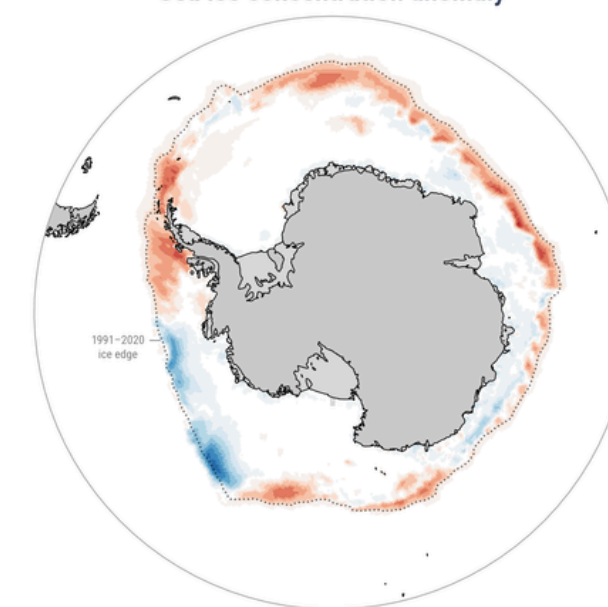
**Antarctic sea ice cover • July 2026**

Data: ERA5 • Reference period: 1991–2020 • Credit: C3S/ECMWF

Sea ice concentration



Sea ice concentration anomaly

PROGRAMME OF
THE EUROPEAN UNION

IMPLEMENTED BY



REFERENCES

1

IMO POLARIS

<https://www.imo.org/>

2

Wagner et al. (2017)

[https://arxiv.org/abs/
1610.06403](https://arxiv.org/abs/1610.06403)

3

Ivanova et al. (2015)

[https://tc.copernicus.org/
articles/9/1797/2015/](https://tc.copernicus.org/articles/9/1797/2015/)

4

Lavergne et al. (2019)

[https://tc.copernicus.org/
articles/13/49/2019/](https://tc.copernicus.org/articles/13/49/2019/)

5

USNIC Antarctic Icebergs

[https://usicecenter.gov/
Products/AntarcticIcebergs](https://usicecenter.gov/Products/AntarcticIcebergs)